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A Robust Method for Gaussian Profile Estimation in the Case of Overlapping Objects

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ABSTRACT The precise stellar object identification is one of the major research fields in astronomy. In astronomical images, the 2D Gaussian function provides a good approximation of stellar objects. This paper proposes a robust method for the estimation of the 2D Gaussian profiles in astronomical images when the stellar components are overlapped. The proposed method efficiently separates the regions of interest of close components without knowing neither their exact number nor accurate center positions. Therefore, fainter objects can be identified, and outliers can be detected. The proposed method is an iteratively reweighted least squares method where weights are updated in each step using a robust measure of residual error dispersion. Soft thresholding based on robust statistics increases the accuracy of the estimated 2D profile parameters and efficiently solves the problem of outliers. The comparison of the proposed method with the conventional least squares method showed a significant modeling gain for one particular set of contributing factors; the difference of magnitudes of close objects, their center-to-center distance, and the widths of their semiaxes relative to the imaging system resolution.

INDEX TERMS Gaussian profile estimation, robust statistics, Huber weights, iteratively reweighted least squares, stellar object identification, overlapping objects.

I. INTRODUCTION

This paper proposes a robust method for fitting a two-dimensional Gaussian profile in the case of overlapping objects. An accurate estimation of Gaussian profile parameters is essential in many scientific and engineering fields. A one-dimensional Gaussian profile fitting has an application in spectrometry to model spectra of emission and absorption lines [1]. The fitting of the two-dimensional Gaussian profile is frequently used in astrometry [2], bioimaging [3], Gaussian beam characterization [4], and different image processing fields where captured images consist of point sources, spatially spread in the image plane. This spread occurs due to lens diffraction in every optical system, even in a theoretically ideal optical system. Moreover, the generalized Gaussian distribution has even more extensive applicability [5], [6].

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The motivation for the use of Gaussian profile estimation in this paper is related to its application in astrometry for the identification of stellar objects from digital images, especially in the case of images with a high spatial density of stars. In other words, stellar objects are point sources spread in the image plane. The spread (blurring) is modeled by the point-spread function (PSF). Although the exact geometrical optical modeling of PSF that considers all possible sources of the spread is extremely complicated even for an objective with a single lens [7], [8], a simplified approximation based on two-dimensional Gaussian function can be used, which is also justified by the central limit theorem. Parameters of the model are different across the image field to account for the anisotropic PSF behavior of the lens. Typically, the spread is more prominent at the frame edges than in the center of the image. Even for the perfectly focused ideal lens, its shape (circular aperture) causes the occurrence of circular PSF, also known as the Airy disk. For real focused lenses with aberrations, PSF is typically circular in the center of the image frame and becomes elliptical and asymmetric at the edges due to the effects of individual aberrations. Parameters of the PSF model vary smoothly across the image field since the spatial PSF evolution, defined by the actual properties of the optical system, is gradual as well. The model is typically estimated as a part of imaging system calibration by estimation and separate modeling of each contributing factor. However, since some of these factors are also time-dependent (e.g., atmospheric dispersion), a better approach would be to estimate the composite model directly from the captured astronomic image itself, without the need for prior calibration. Hence, the anisotropic model is constructed from the estimated PSFs of individual point sources that are contained in the image utilizing spatially localized averaging. As a part of image restoration, in particular image deblurring, the estimated and locally smoothed PSF model also enables the application of deconvolution algorithms that require the knowledge of PSF, such as the Lucy-Richardson algorithm [9]. If the estimated PSF parameters of an individual object deviate significantly from the averaged estimates in the same part of the frame, it might indicate a different kind of light source. Such sources can be various distributed astronomical objects such as galaxies that are elongated, but also measurement errors (e.g., faulty pixels) or noise (e.g., salt and pepper noise). Hence, such deviations in estimated parameters can be used for outlier detection during stellar object identification. However, the basic prerequisite, for all these procedures to work, is an accurate PSF model estimation of each point source contained in the image.

In astronomical images, the main problems in estimating 2D Gaussian parameters are noisy measurements and overlapping objects. Many methods have already been proposed to estimate 2D Gaussian profiles from noisy data. Not many methods that would estimate 2D Gaussian parameters from overlapped profiles by considering the overlap as a kind of stochastic contamination have been proposed so far. The problem of the overlapping objects can be treated as a special case of heavy-tailed noise contamination, but in this case, the error is deterministic rather than stochastic. In both cases, however, the modeling residual represents an outlier. There are two approaches to outliers treatment. The first employs a complete discard of these data before the estimation process, which results in the loss of useful information. The second is the application of robust techniques that reduce the weight of outliers consequently reducing their influence on the final model estimate. The proposed method treats the outliers using robust statistics for residual error analysis and the estimation weights update.

Conventionally, the parameter estimation assumes the optimization process that searches for an optimal solution in the least squares (LS) sense or in the maximum likelihood (MLE) sense. The LS estimators are widely used in many industrial fields [10], [11]. The MLE and LS estimates will be the same if the image is corrupted only with the additive Gaussian noise. However, no method gives accurate estimations in the presence of heavy-tailed distribution noise (e.g., caused by the close object contamination). Therefore, a robust iteratively reweighted least squares (IRWLS) algorithm is proposed. It gives a more accurate estimate of the actual 2D Gaussian profile parameters in the case of properly constructed weights that converge to the solution that optimally deweights unwanted overlapping objects. The goal was to assess the conditions in which the problem of overlapping objects can be solved using such a robust method for the ideal case without the additional noise.

II. RELATED WORK

The underlying mathematical formulation for estimating 2D Gaussian parameters represents the solution of the overdetermined system of nonlinear equations. One solution is to use an iterative procedure such as the Newton-Raphson method. The method does not guarantee convergence, and the result of the optimization process is highly dependent on the selection of initial parameters [12].

Moment-based methods are accurate for determining the center of a Gaussian centroid with subpixel accuracy but do not estimate other parameters [13].

In [14], the process of 1D Gaussian profile estimation is accelerated because the 1D Gaussian function is an exponential of a quadratic function. The natural-logged data were fitted in the sense of least squares into the logarithm of the Gaussian function, thus yielding a system of linear equations with the analytical solution. The disadvantage of this approach is found in the fact that the impact of noise present in the observed data was not considered. In [15], the influence of noise on the estimation accuracy of 1D Gaussian profile parameters was analyzed. It was proved that the influence of noise in the natural-logged data can be significant, especially if the signal values are initially small. To reduce the influence of small data values, the method of weighted least squares was applied, which set the weights based on actual data values. Furthermore, the iterative procedure solved the problem of long-tailed Gaussian function samples in the observed data. In [16], the method based on polynomials was introduced, wherein the partial derivatives were used instead of a natural logarithm. A fast two-step algorithm was proposed in [17]. In the first step, the area under the Gaussian function was used for the direct determination of the profile width. In the second step, the weighted least squares method was used to estimate the magnitude and mean value, and the weights were calculated as squared data values.

Another approach, used in [18] and [19], was to search for MLE estimates of 1D and 2D Gaussian profile parameters in the case of Gaussian noise contamination. MLE solution gives minimum-variance unbiased estimates in the case of a large number of measurement samples, i.e., it is more efficient than the LS method. Analytical expressions for derivations and for the Hessian matrix of negative log-likelihood are also given in [18], [19] to facilitate the procedure of finding the optimal estimate. Cramer-Rao bound was derived and used to obtain analytical expressions for the expected parameter variances, thus avoiding the use of the numerical or Monte Carlo method for a parameter variance estimation.

As far as we know, this is the first method that deals with the estimation of 2D Gaussian profile parameters in the case of contamination by another close object.

III. MOTIVATION

Overlapping star components complicate the detection of stellar objects and the precise determination of their PSFs' parameters. The main problem is how to determine the affiliation of pixels to a single source. Commonly used methods for spatial segmentation assume the presence of a local minimum between the components. Commonly used methods for spatial segmentation, such as centroidal Voronoi tessellation, fail when the local minima between the points vanish since the close point sources exceed the resolution limit defined by the Sparrow criterion and its modified version [20], [21]. These well-known criteria determine the resolution limit of two equally and unequally bright points for diffraction-limited optics, thus assuming that a point-source PSF is the ideal Airy disk. The criteria are derived from the conditions for the occurrence of the inflection point between two close Airy disks determined by their center-to-center distance and difference in brightness. However, the PSF profiles of stellar components in actual images are far from the ideal Airy disks due to the variety of aberrations that introduce unavoidable degradations in the imaging process. In our approach, these degradations are described by the 2D Gaussian function. Although different resolution criteria exist (e.g., Rayleigh criterion, Dawes' limit [22]), these are inapplicable for a strictly positive Gaussian function. However, the Sparrow inflection criterion can be derived analytically. Without the loss of generality, we assume the simple example of two overlapped 1D Gaussian profiles of the common width σ that are placed at $x = \mu_1 = 0$ for the first stronger component and $x = \mu_2 = d$ for the second weaker component. The total sensed flux can be written as

$$f(x) = \exp\left(\frac{-x^2}{2\sigma^2}\right) + A\exp\left(-\frac{(x-d)^2}{2\sigma^2}\right), \; A \leq 1. \quad (1)$$

By substituting $x = \sqrt{2}\sigma\xi$ and $d = k\sigma$ in (1), it becomes

$$f(\xi) = \exp(-\xi^2) + A\exp\left(-\left(\xi - \frac{k}{\sqrt{2}}\right)^2\right). \quad (2)$$

The inflection point occurs when the first and the second derivatives vanish at the same point $x = x_0$:

$$\left.\frac{\delta f(x)}{\delta x}\right|_{x=x_0} = \frac{\delta f(x)}{\delta \xi}\frac{\delta \xi}{\delta x} = 0, \quad (3)$$

$$\left.\frac{\delta^2 f(x)}{\delta x^2}\right|_{x=x_0} = \frac{\delta^2 f(x)}{\delta \xi^2}\frac{\delta^2 \xi}{\delta x^2} = 0. \quad (4)$$

For the two components displaced by $d = k\sigma$, the obtained inflection point x_0 and the corresponding amplitude A of the weaker component are determined by the chosen

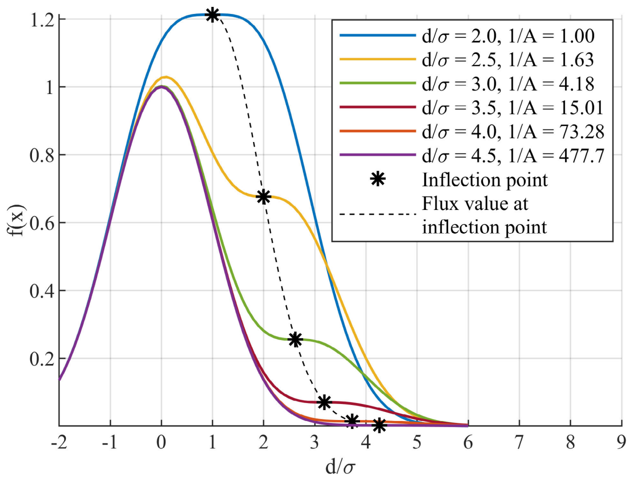


FIGURE 1. Sums of two 1D Gaussian profiles with parameters: $\sigma = 1$, $\mu_1 = 0$ and $\mu_2 = d$ for different center-to-center distances d/σ and ratios of their peak amplitudes $1/A$.

value of k as

$$x_0 = \frac{\sigma}{2}(k + \sqrt{(k^2 - 4)}), \quad A \leq 1, \quad (5)$$

$$A = \frac{\exp^{(-\frac{x_0}{\sqrt{2}\sigma})^2}}{\exp^{-(\frac{x_0}{\sqrt{2}\sigma} - \frac{k}{\sqrt{2}})^2}}\left(\frac{-\frac{x_0}{\sqrt{2}\sigma}}{\frac{x_0}{\sqrt{2}\sigma} - \frac{k}{\sqrt{2}}}\right). \quad (6)$$

For the trivial case of two equal components ($A = 1$), the inflection point occurs at midpoint $x_0 = \sigma$ when components are displaced by $d = 2\sigma$, which corresponds to the minimum value of $k = 2$. For $k = 2.5$ and the corresponding displacement $d = 2.5\sigma$, the inflection point is at $x_0 = 2\sigma$ and the amplitude ratio is $1/A = 1.63$. Generally, for $A < 1$ the inflection point is always closer to the weaker component, and the separation is always greater than $d = 2\sigma$, i.e., $k > 2$. The inflection point determines the resolution upper limit. Conventional methods based on the local minimum/maximum detection can not separate Gaussian components placed closer than the resolution limit. Our motivation is to introduce the method that would show the improvement even when close components violate the resolution limit. Fig. 1 shows the family of curves for six combinations of center-to-center distances d/σ and ratios of their peak amplitudes $1/A$. Conditions (3) and (4) can be directly generalized for the multi-dimensional case assuming rotationally symmetric Gaussians of the same width in all dimensions.

IV. PROPOSED METHOD

This section gives an overview of the proposed method. Subsection A. describes the model of an astronomical image as a set of atoms. These atoms are the part of the parametric dictionary described in subsection B. Subsections C., D., and E. formulate the problem of atom estimation and compare the conventional LS method with the proposed IRWLS method. Finally, section F. analyzes the method's efficiency.

A. IMAGE MODEL AND ITS ESTIMATION

The estimation of the 2D Gaussian profile parameters begins with the selection of the region of interest (ROI) that involves all pixels enclosed within the profile being estimated, and as few as possible pixels that belong to adjacent objects or background. The high density of stellar objects in astronomical images causes the overlapping of two or more close objects making the ROI selection and the estimation process more difficult.

The detection of stellar objects in astronomical images and determination of their ROIs can be performed in various ways. For the sake of simplicity, the method of local extrema is used in this paper. The found local extrema, sorted by their value in decreasing order, determine the order of the estimation and extraction process. The ROI is always centered at the current position of the local extreme. Thus, expected objects are successively estimated and removed from the image with the purpose of maximal reduction of the remaining energy. The ROI width is adjusted to the expected profile width (e.g., enclosing $\pm 3\sigma$ around the center). The described approach is similar to the conventional matching pursuit method [23]–[29]. The matching pursuit is an iterative greedy algorithm for the sparse approximation of the signal that assumes that the signal can be represented as a linear combination of the atom functions from the complete or over-complete dictionary multiplied by the projection coefficients. The algorithm sequentially selects the best matching atom function from the dictionary by searching for the largest inner product between the signal and dictionary columns, thereby minimizing the residual energy in each iteration.

The astronomical image can be considered as a sparse combination of a constrained number of atoms that are chosen as 2D Gaussians of arbitrary shape and position. Gaussians can be parametrized with their peak positions and corresponding covariance matrices. Since all parameters are real numbers, the size of such a parametric dictionary is infinite. By selecting the region of interest (ROI), the search area of the infinite parametric dictionary is narrowed to a subset of atoms. Although the search area is spatially localized, even this subset is still of the infinite dimension. The sequential selection and removal of the estimated objects require accurate parameter estimation to avoid the impact of the erroneous object removal to the next iterations. For the given ROI, the chosen atom and its projection coefficient are found through nonlinear optimization over its parameters. The application of the LS criterion in the optimization process causes significant estimation error for overlapping objects since the chosen atom is trying to represent both objects with a single Gaussian. Inasmuch as the shape mismatch irretrievably compromises the residual, all subsequent iterations are affected. On the other hand, by applying the robust metrics, soft deweighting of the weaker components can be achieved so that the chosen optimal atom represents only the strongest component of the sum, while the weaker components are in the ideal case modeled and removed in the subsequent iterations.

Most interestingly, applying the robust method ultimately yields a smaller squared error than the LS method due to the greediness of the LS criterion. The LS method has a smaller squared error in a single iteration but yields worse results in the long run as it will be shown in several examples and experiments.

B. PARAMETRIC DICTIONARY

The mathematical form of the 2D Gaussian profile atom can be written as

$$f(\mathbf{v}_i) = \frac{A_l}{2\pi\sqrt{\det(\mathbf{C})}} \exp\Big[-\frac{1}{2}\Big((\mathbf{v}_i-\boldsymbol{\mu})^T\mathbf{C}^{-1}(\mathbf{v}_i-\boldsymbol{\mu})\Big)\Big], \tag{7}$$

where $\mathbf{v}_i = [x_i \;\; y_i]^T$ in the quadratic form [1] denotes a sample position of the ith pixel with the readout value z_i that is contained within the ROI of the profile being estimated. The peak position of the profile atom is denoted with $\boldsymbol{\mu} = [\mu_x \;\; \mu_y]^T$, while the corresponding covariance matrix that describes the shape of the atom is denoted with $\mathbf{C}$. Finally, A_l denotes a linear amplitude scale that corresponds to the projection coefficient of the atom.

The readout pixel value z_i is linearly proportional to the part of the radiant flux of the astronomical object that impinges on the ith pixel. Mathematically, the readout value is proportional to the integral of the product of the radiation source spectral density; of the overall spectral response of the optical system including the input filter and the spectral response of the sensor that is spatially integrated within the active pixel area. The image of the selected astronomical object is projected onto the sensor plane, activating the finite number of neighboring pixels within the ROI. By using a normalized Gaussian representation in (7), the linear scale A_l represents the total sensed radiant flux, i.e., the intensity of the observed point source, invariant of the PSF shape or width. Various PSFs only cause a different spatial spread of the same flux. The total flux represented by A_l is proportional to the number of photons detected in the sensor plane in the unit time in all pixels enclosed within the PSF.

For astronomical imaging, the dynamic range of such filtered flux is huge, so instead of linear representation, it is usually represented in the logarithmic domain. In astrophotometry, the conventional base of this logarithmic representation is chosen as $\sqrt[5]{100}$, which is also known as the magnitude scale. Therefore, the linear amplitude scale A_l can be converted to the negative[2] magnitude scale A_m as $A_m = \log_{\sqrt[5]{100}} A_l$. The difference of magnitudes of two stellar objects ΔA describes the difference in their brightness. For the example of two objects with a magnitude difference $\Delta A = 5$, the equivalent ratio of their linear amplitudes is

[1] The quadratic form in the argument of the Gaussian function represents the squared Mahalanobis distance of the samples from the peak position. By choosing the ROI within $k\sigma$ around the peak position, we are defining a convex input domain with the radius of the Mahalanobis distance equal to k in all directions.

[2] The magnitude scale is reverse logarithmic so the brighter the object, the lower the magnitude.

1/100. Such a logarithmic representation also facilitates the numerical stability of the optimization process and is used in further expressions.

In the uncorrelated form, the covariance matrix can be calculated as a quadratic form of an orthonormal rotation matrix $\mathbf{R}$ and a diagonal matrix of eigenvalues $\mathbf{S}$:

$$\mathbf{R} = \begin{bmatrix} cos(\frac{\rho}{2}) & -sin(\frac{\rho}{2}) \\ sin(\frac{\rho}{2}) & cos(\frac{\rho}{2}) \end{bmatrix}, \quad \mathbf{S} = \begin{bmatrix} \lambda_1^2 & 0 \\ 0 & \lambda_2^2 \end{bmatrix}, \ \mathbf{C} = \mathbf{RSR}',$$

where λ_1 and λ_2 represent standard deviations in directions of the major and minor semiaxes, while $\frac{\rho}{2} = \theta$ is the rotation angle. The double rotation angle ρ is introduced as an auxiliary optimization variable to ensure the continuity of the parameter at the boundary between $\frac{\rho}{2} = \pi/2$ and $\frac{\rho}{2} = -\pi/2$, which are shape-equivalent and represent the same atom.

Consequently, the atom parameters spanning the dictionary are the peak position (μ_x, μ_y), the widths of semiaxes λ_1, λ_2, and the double rotation angle ρ, while the projection coefficient is represented by the negative magnitude scale A_m. Thus, the parameters' vector for the optimization process becomes

$$\boldsymbol{\beta} = [A_m \ \lambda_1 \ \lambda_2 \ \mu_x \ \mu_y \ \rho]. \tag{8}$$

The estimation process involves optimization over all samples enclosed within the determined ROI $((x_i, y_i) \in ROI)$. For the pixel value z_i and its position $\mathbf{v}_i = [x_i \ y_i]$, the pixel error of the model $f(\mathbf{v}_i, \boldsymbol{\beta})$ is

$$e_i = z_i - f(\mathbf{v}_i, \boldsymbol{\beta}). \tag{9}$$

C. LS AND IRWLS METHODS

The LS method minimizes the squared residual error given as

$$D(\boldsymbol{\beta}) = \sum_{i \in ROI} w_i(e_i) e_i^2, \tag{10}$$

with unit weights w_i for all samples within the profile being estimated. It is a greedy method that will, in the case of overlapping object contamination, search for the compromise solution, currently optimal in the LS sense. The obtained solution generally does not correspond to the actual parameters of any of the two or more overlapping components. Therefore, an iteratively reweighted least squares method (IRWLS) that minimizes the weighted sum of least squares is introduced. Ideally, it recognizes the samples belonging to other objects as outliers and reduces their weights, thus decreasing their contribution to the final estimate. Although such weights can be determined in a variety of ways, the application of Huber weights [30] in the IRWLS method represents soft thresholding that is solely based on a statistical analysis of the residual error.

D. PROBLEM FORMULATION

The IRWLS method iteratively solves the following problem

$$\hat{\boldsymbol{\beta}} = \arg\min_{\boldsymbol{\beta}} D(\boldsymbol{\beta}). \tag{11}$$

The weights in the current step are constructed according to the robust analysis of the residual error from the previous step. The weights are constructed as Huber weights, as follows

$$w_i(e_i) = w_H(e_i) = \begin{cases} 1, & \text{if } |e_i| \le \delta \\ \delta/|e_i|, & \text{if } |e_i| > \delta, \end{cases} \tag{12}$$

where e_i denotes the residual error of the ith pixel, and δ denotes a threshold correlated with robust estimation of standard deviation SD of residual errors. The Huber estimator combines L_1 and L_2 norm criteria. The L_1 norm is applied to outliers, while the L_2 norm is applied to regular samples. Therefore, the samples with the absolute error within the threshold δ have unit weights (L_2), while the weights of outliers are decreased to mimic the L_1 norm. The smaller threshold δ emphasizes the L_1 norm, making the estimator resilient to outliers but asymptotically less efficient. The larger value of δ increases the influence of the L_2 norm and makes the estimator more efficient, but also more sensitive to outliers. In our experiments, the threshold δ was selected as $\delta = 1.345SD$ such that the Huber estimator still has sufficiently high asymptotic efficiency (95%) when the errors are normally distributed [31]. The standard deviation SD of residuals of all pixels within the ROI was constructed using a robust measure of the error dispersion. One such measure is $\hat{SD} = MAD/0.6745$, where MAD denotes the median of the absolute deviations from the data's median. Inter-quartile range or other robust measures of dispersion can also be used alternatively. The IRWLS method, which uses the robust statistics for the estimation of error dispersion, can be generally applied to the estimation of model parameters from the observed data corrupted with outliers or heavy-tailed noise. However, the prerequisite for the application of robust statistics is the presence of at least 50% of valid samples in the observed data. Additionally, the assumption is that the histogram of errors is bi-modal with separable modes. The pseudocode of the IRWLS method is given in Algorithm 1. It consists of two nested loops. The inner loop solves the equation (11) in step 9 of Algorithm 1 through nonlinear optimization for fixed weights computed from residuals of the previous iteration, while the outer loop modifies the weights for the subsequent iteration.

An optimization process can be performed even without analytical gradients by calculating the numerical gradients of the objective function. In order to accelerate the optimization step 9, the partial derivatives of (10) for fixed weights are derived as

$$\frac{\delta D}{\delta A_m} = \frac{\delta D}{\delta A_l}\frac{\delta A_l}{\delta A_m} = \ln \sqrt[5]{100} \sum_{i=1}^{n} \tau_i,$$

$$\frac{\delta D}{\delta \mu_x} = \sum_{i=1}^{n} \Big(\frac{\tau_i}{2\lambda_1^2 \cdot \lambda_2^2} \Big((x_i - \mu_x) \cdot ((\lambda_1^2 + \lambda_2^2) + \cos(\rho) \cdot (\lambda_2^2 - \lambda_1^2)) + (y_i - \mu_y) \cdot ((\lambda_2^2 - \lambda_1^2) \cdot \sin(\rho)) \Big) \Big),$$

Algorithm 1 IRWLS Method for Single ROI

Input: positions of all n samples within the ROI and their values $(\mathbf{v}_i, z_i) \in ROI$

Output: parameters $\hat{\boldsymbol{\beta}}$

1: $t \leftarrow 0$
2: $w_i^{(0)} \leftarrow 1, \ \forall i \in ROI$
3: $\hat{\boldsymbol{\beta}}^{(0)} \leftarrow \{A, \lambda_1, \lambda_2, \mu_x, \mu_y, \rho\}$ ▷ initial LS estimates
4: $t \leftarrow t + 1$
5: **do** ▷ outer loop
6: $\quad e_i^{(t-1)} \leftarrow z_i - f(\mathbf{v}_i, \hat{\boldsymbol{\beta}}^{(t-1)}), \ \forall i \in ROI$
7: $\quad \delta^{(t)} = 1.345 \cdot MAD(\mathbf{e}^{(t-1)})/0.6745$
8: $\quad w_i^{(t)} \leftarrow w_H(e_i^{(t-1)}, \delta^{(t)}), \ \forall i \in ROI$ ▷ (12)
9: $\quad \hat{\boldsymbol{\beta}}^{(t)} \leftarrow \arg\min_{\boldsymbol{\beta}} D(\boldsymbol{\beta})$ ▷ inner loop (11)
10: $\quad t \leftarrow t + 1$ ▷ outer loop iteration
11: **while** $\frac{|\delta^{(t-1)} - \delta^{(t)}|}{\delta^{(t-1)}} > tol \ \ \& \ \ t < t_{max}$

$$\frac{\delta D}{\delta \mu_y} = \sum_{i=1}^{n} \Big(\frac{\tau_i}{2\lambda_1^2 \cdot \lambda_2^2} \Big((y_i - \mu_y) \cdot ((\lambda_1^2 + \lambda_2^2) + \cos(\rho) \cdot (\lambda_1^2 - \lambda_2^2)) + (x_i - \mu_x) \cdot ((\lambda_2^2 - \lambda_1^2) \cdot \sin(\rho)) \Big) \Big),$$

$$\frac{\delta D}{\delta \rho} = \frac{\delta D}{\delta \theta} \frac{\delta \theta}{\delta \rho} = \frac{1}{2} \sum_{i=1}^{n} \Big(\tau_i (\lambda_1^2 - \lambda_2^2)/(\lambda_1^2 \cdot \lambda_2^2) \cdot \Big((y_i - \mu_y) \cdot \cos(\frac{\rho}{2}) - (x_i - \mu_x) \cdot \sin(\frac{\rho}{2}) \Big) \cdot \Big((y_i - \mu_y) \cdot \sin(\frac{\rho}{2}) + (x_i - \mu_x) \cdot \cos(\frac{\rho}{2}) \Big) \Big),$$

$$\frac{\delta D}{\delta \lambda_1} = \sum_{i=1}^{n} \Big(\tau_i \Big((x_i - \mu_x)^2 \cdot \cos(\frac{\rho}{2})^2 + (y_i - \mu_y)^2 \cdot \sin(\frac{\rho}{2})^2 + \sin(\rho) \cdot (y_i - \mu_y) \cdot (x_i - \mu_x) - \lambda_1^2 \Big) / \lambda_1^3 \Big),$$

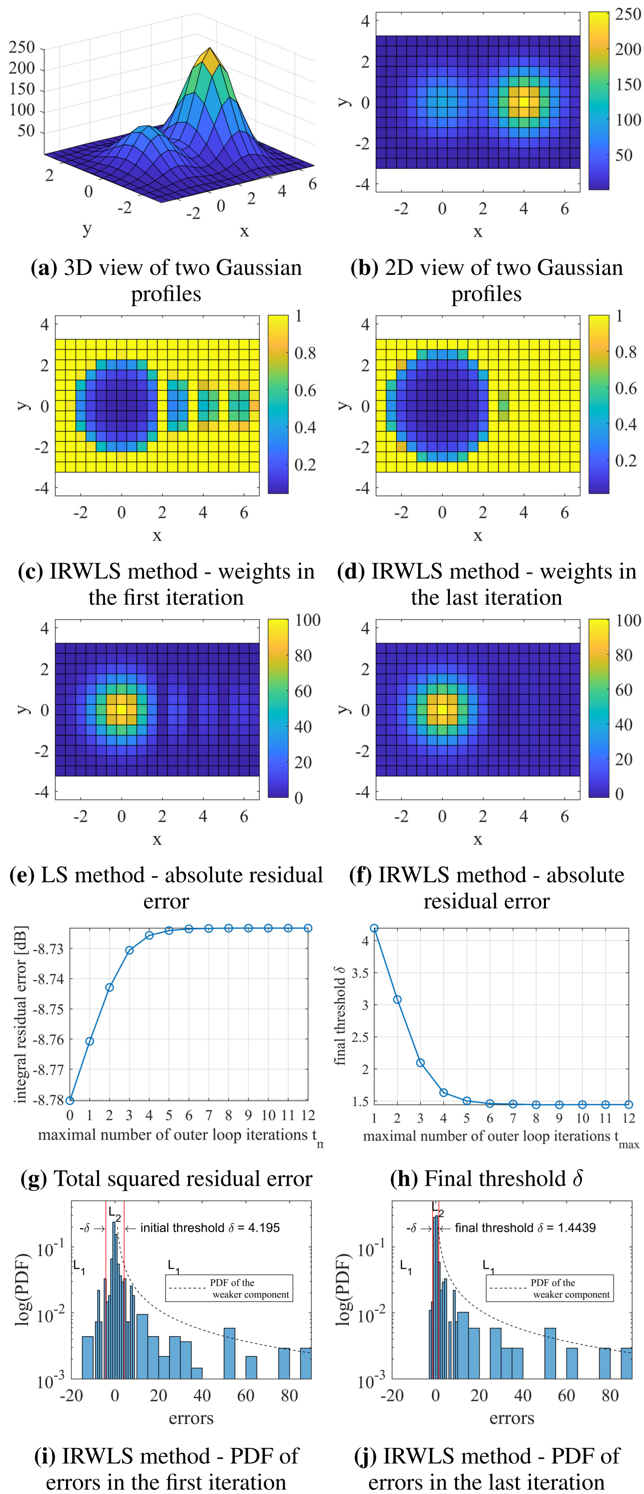
$$\frac{\delta D}{\delta \lambda_2} = \sum_{i=1}^{n} \Big(\tau_i \Big((x_i - \mu_x)^2 \cdot \sin(\frac{\rho}{2})^2 + (y_i - \mu_y)^2 \cdot \cos(\frac{\rho}{2})^2 - \sin(\rho) \cdot (y_i - \mu_y) \cdot (x_i - \mu_x) - \lambda_2^2 \Big) / \lambda_2^3 \Big),$$

where $\tau_i = 2 \ w_i(f(\mathbf{v}_i, \boldsymbol{\beta}) - z_i) f(\mathbf{v}_i, \boldsymbol{\beta})$ is the weighted correlation between the error and the model.

As in any optimization process, the finding of an optimal solution of step 9 in Algorithm 1 also requires an appropriate initial guess of model parameters that should be sufficiently close to the optimal solution. Various techniques can be used to compute the initial model (the method of local extrema, the first and the second moments, the centroid method).

E. MODEL ESTIMATION

To illustrate the method, let us take a simple synthetic example of two symmetric 2D Gaussian profiles, with the same widths of semiaxes ($\lambda_1 = \lambda_2 = \sigma = 1$), but with different magnitudes ($\Delta A = 1$), and with the center-to-center distance $d = 4\sigma$ (Fig. 2a and Fig. 2b). The image with two components is synthesized with vectors of parameters: $\boldsymbol{\beta}_1 = [A_m, \lambda_1, \lambda_2, \mu_x, \mu_y, \rho] = [8, 1, 1, 4, 0, 0]$ and



(a) 3D view of two Gaussian profiles
(b) 2D view of two Gaussian profiles
(c) IRWLS method - weights in the first iteration
(d) IRWLS method - weights in the last iteration
(e) LS method - absolute residual error
(f) IRWLS method - absolute residual error
(g) Total squared residual error
(h) Final threshold δ
(i) IRWLS method - PDF of errors in the first iteration
(j) IRWLS method - PDF of errors in the last iteration

FIGURE 2. Estimation of the stronger 2D Gaussian profile in the case of two overlapping components using the IRWLS method with uniform sampling grid and the pixel size of 0.5 in both directions. The total number of outer loop iterations is 12.

$\boldsymbol{\beta}_2 = [7, 1, 1, 0, 0, 0]$. The ROI is chosen as a rectangular region enclosing both profiles exactly as it is shown in Fig. 2. To circumvent the issue of initialization for model solving, the initial model supplied to the optimization algorithm was chosen to be identical to the specified models $\boldsymbol{\beta}_1$ and $\boldsymbol{\beta}_2$.

Without overlapping, this would also be the optimal solution, but the overlapping profiles cause divergences from the specified optimal point.

1) ESTIMATION OF THE STRONGER COMPONENT

Firstly, the stronger component is estimated using the estimation weights w_i that are shown in Fig. 2c and Fig. 2d. The weights are changed in each outer loop iteration of Algorithm 1. In the first iteration, the weights are obtained from the residuals of the LS solution.

The effectiveness of the IRWLS method comes to the fore by increasing the number of outer loop iterations. The unit weights are gradually assigned to the stronger component, while the weights close to zero are assigned to the weaker component. It means that the IRWLS method successfully focuses on the stronger component and recognizes the samples of the weaker component as outliers.

2) STATISTICAL ANALYSIS OF RESIDUAL ERRORS

Fig. 2e and Fig. 2f show the residual errors e_i (9) of the LS and IRWLS methods after the removal of the estimated stronger component. Fig. 2g shows that, after the removal of the stronger component, the squared residual error increases in each iteration of the outer loop with the smallest value in iteration zero which corresponds to the LS solution. Such an increase of the total squared residual error is a natural behavior of the IRWLS algorithm since the weighted error is reduced at the expense of the increased squared error. However, it will be shown that the IRWLS method will yield a smaller total residual error after the modeling and removal of both components. The rate of convergence of the IRWLS method can be observed from the change of the threshold δ as it is shown in Fig. 2h since the weights are determined solely on δ and e_i from the previous iteration. The threshold δ achieves the stationary state after approximately six outer loop iterations and further iterations do not significantly increase the modeling gain, as it will be shown after the removal of the second component. In this example, the threshold δ in the estimation of the stronger component achieves the value of 1.44, and the outer loop terminates after 12 iterations since the relative change of the threshold δ is smaller than the given tolerance $tol = 10^{-5}$. The estimated parameters' vector of the stronger component by using the LS method is $\hat{\boldsymbol{\beta}}_1 =$ [8.0569, 1.0912, 1.0000, 3.9557, 0.0000, 0.0000], while the IRWLS method yields $\hat{\boldsymbol{\beta}}_1 =$ [8.0122, 1.0145, 1.0001, 3.9885, 0.0000, 0.0000]. As expected, the LS solution gives the elongated profile along the x axis with $\lambda_1/\lambda_2 = 1.09$ with the center of the profile shifted by 0.045 from its original position, while the IRWLS solution is less elongated and with almost four times smaller center shift. The IRWLS-based solution with Huber weights gives a desirable residual error in which the stronger profile is removed almost completely, while the weaker one is left more or less untouched.

Fig. 2i and Fig. 2j show the log probability density function (PDF) of residual errors for the first and the last iteration of the IRWLS method approximated from its histograms. The bin widths of these histograms were adapted to the error statistics, i.e., the narrow bins of the unit width were used for small errors while wider bins of width five were used for large residuals. The threshold δ, which is also shown in these two figures, represents the action limit of the L_1 and L_2 norms. It is decreased in each IRWLS iteration since the estimation model gradually adapts to the stronger component and deweights the weaker one. Consequently, the method spatially discriminates against the profiles' support by solely using the information of the residual errors. The residual errors in the last iteration are mainly positive. In the case of two overlapping components, the assumption is that these positive residuals are comprised of the original samples of the weaker component. Besides the obvious visual matching of the residual with the second component (Fig. 2f), the further proof would be the matching of the PDF of the weaker component with the PDF of errors while taking into account the finite sample size due to the discrete nature of the regular 2D profile sampling.

Considering a continuous rotationally symmetric 2D Gaussian profile of the width σ defined over the finite support $\pm k\sigma$ around its mean, the PDF of its values can be derived from the assumed PDF of its argument. Samples of the 2D Gaussian profile are calculated as a function of two-dimensional random variable $\mathbf{v} = [x, y]$ uniformly distributed within the profile support. If the Gaussian profile is described as $z(\mathbf{v}) = A_l \exp(\frac{-1}{2}\frac{\|\mathbf{v}\|_2^2}{\sigma^2})$ and if the argument vector $\mathbf{v}$ is uniformly distributed within the circular region of radius $k\sigma$ as

$$f_v(\mathbf{v}) = \begin{cases} \dfrac{1}{(k\sigma)^2\pi}, & \text{if } \|\mathbf{v}\|_2^2 \le k^2\sigma^2 \\ 0, & \text{otherwise,} \end{cases} \tag{13}$$

then the PDF of z is readily found as

$$f_z(z) = \begin{cases} \dfrac{2}{k^2\, z}, & \text{if } A_l \exp(\dfrac{-k^2}{2}) \le z \le A_l \\ 0, & \text{otherwise.} \end{cases} \tag{14}$$

Such hyperbolic PDF is also known as the log-uniform distribution. For the actual discrete case, the profile is evaluated over a regular grid resulting in the finite number of pixel values within the circular region. However, this can be considered as a realization of a stochastic process drawn from the derived continuous distribution, so the empirical distribution approximated from the histogram of z_i shall correspond to the continuous PDF.

The expected continuous PDF of the weaker component according to (14) is shown with dashed lines in Fig. 2i and Fig. 2j. The histograms of errors over finite bins follow this PDF, thereby confirming the assumption that positive samples belong to the weaker untouched component. Naturally, this is only true if the profile of the weaker component is fully enclosed within the ROI. Since the log-uniform distribution exists only above the bounded support, it cannot fulfill the formal condition for a heavy-tailed distribution. Nevertheless, log-uniform distributed errors arising from the weaker component with hyperbolic PDF can still be considered as

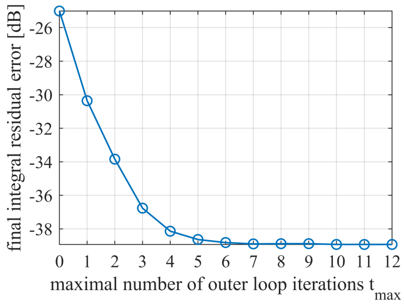


FIGURE 3. **Final squared residual error after removal of both profiles for the bounded number of outer loop iterations t_{max}. Iteration zero represents the LS solution.**

heavy-tailed contamination compared to the much narrower normally distributed modeling errors of the stronger component, thus enabling the separation of the errors from the mixture of distributions by a suitably chosen threshold δ. This example shows the high effectiveness of the IRWLS method in solving the problem of profile overlapping by considering its effect as heavy-tailed contamination.

3) ESTIMATION OF THE WEAKER COMPONENT

Once the stronger profile is removed, the estimation of the weaker component is performed. The estimated parameters of the weaker component using the LS method are $\hat{\boldsymbol{\beta}}_2 = [6.9303, 1.0000, 0.9193, -0.0578, 0.0000, -3.1416]$, while the IRWLS method yields $\hat{\boldsymbol{\beta}}_2 = [6.9913, 1.0002, 0.9914, -0.0089, 0.0000, -3.1416]$. The threshold δ in the estimation of the weaker component achieves the value of 0.15 and the outer loop terminates after only 5 iterations due to the achieved relative tolerance $tol = 10^{-5}$. Again, the IRWLS solution is much closer to the specified model $\boldsymbol{\beta}_2$ than the LS solution with much smaller parameters' error. It is obvious that the IRWLS method yields more accurate parameters than the LS method for both components.

4) TOTAL MODELING GAIN

After the removal of both estimated components, the final squared residual error can be used as a composite fitness measure instead of the parameters' errors because such a simple scalar measure indirectly demonstrates the estimation accuracy of each individual parameter. The final squared residual error after the removal of all estimated objects defines the total modeling gain

$$g = -10\log_{10}\left(\frac{\sum_{\forall i\in I}\left(z_i - \sum_{j=1}^{M} f(\mathbf{v}_i, \hat{\boldsymbol{\beta}}_j)\right)^2}{\sum_{\forall i\in I} z_i^2}\right), \quad (15)$$

where z_i represents the ith pixel value in the image I, while $f(\mathbf{v}_i, \hat{\boldsymbol{\beta}}_j)$ represents the estimated model for a single profile, and M is the number of modeled profiles in the image.

The total modeling error for two objects is -25.0045 dB for the LS method and -38.9272 dB for the IRWLS method after 12 iterations for the first component and 5 iterations for the second (Fig. 3). The outer loop is terminated by either achieving convergence tol, defined by the relative change of the threshold δ, or by exceeding the maximum number of iterations t_{max}. To investigate the convergence rate of the IRWLS method, the maximum number of outer loop iterations t_{max} was bounded from 0 (simple LS solution) up to maximum 12 and the total resulting modeling error for both components was evaluated for each case, as shown in Fig. 3. It can be seen that 90% of the modeling improvement is achieved after four iterations of the loop.

F. METHOD EFFICIENCY

It can be assumed that the efficiency of the IRWLS method is primarily influenced by three contributing factors: a) center-to-center distance of objects d, b) difference of magnitudes ΔA, and c) their nominal widths λ_1 and λ_2 relative to the pixel size. Specifically, the absolute difference of magnitudes

$$\Delta A = |A_{m_1} - A_{m_2}|, \quad (16)$$

is sufficient to fully describe the relative relations of objects' brightness for the ideal noiseless case of the overlapping objects. For actual images, such simplification is not valid since the efficiency is also influenced by the noise level relative to the A_{m_1} and A_{m_2}, but our goal in this paper is to determine the efficiency bounds for the ideal noiseless case. There have already been proposed estimation methods in other research fields that ensure the robustness even in the case of noisy data [32], [33], but not for the 2D Gaussian parameter estimation.

The center-to-center distance d of overlapping profiles affects the number of valid pixels that belong to each object, and indirectly, the ability of the method to determine the soft boundaries between the components to separate them.

The ratio of the object width and the pixel size also affects the number of valid pixels used in the estimation process and, consequently, the variance of the estimated parameters.

V. SIMULATIONS AND RESULTS

Since the estimation weights depend on residual error analysis in each step, it is hard to obtain analytical expressions for parameter variances. Therefore, the algorithm efficiency and the parameters' accuracy are verified numerically using a Monte Carlo simulation.

A. ANALYSIS OF THE INFLUENCE OF CONTRIBUTING FACTORS

In the first part of the experiment, the influence of two contributing factors on the modeling gain was analyzed. These parameters were the difference of objects' magnitudes ΔA and their center-to-center distance d for fixed widths of their semiaxes ($\lambda_1 = \lambda_2 = \sigma = 1$ pix for both objects). The Monte Carlo simulation was performed for a set of center-to-center distances d in the range from 0 to 6σ, with the step size of 0.5σ and differences of profiles' magnitudes ΔA in the range from -5 to 5 with the step size of 0.5. The total number of evaluated pairs was 143 due to the inherent symmetry for positive and negative differences of magnitudes. For each chosen pair, one simulation with 192 synthesized sub-images was performed. An example of such an image is shown in Fig. 4. Each sub-image contained two objects

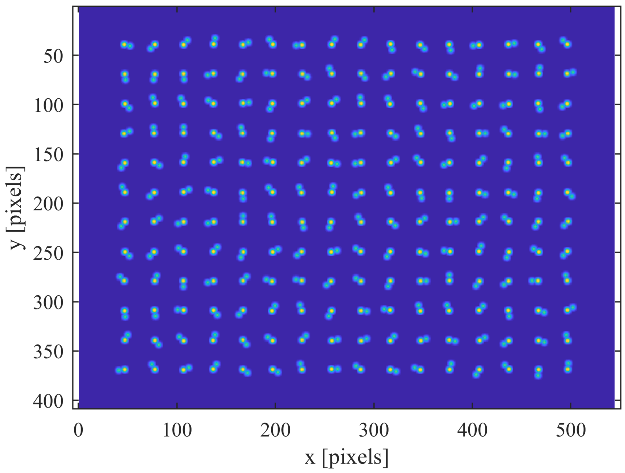
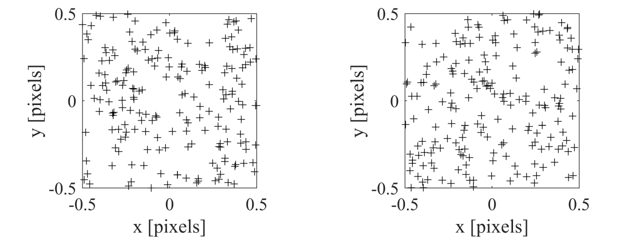


FIGURE 4. **Synthetic image of 192 sub-images and 384 objects. Objects' parameters: $\sigma = 1$ *pix*, $d = 6\sigma$, $\Delta A = 3$.**



(**a**) Random fractional shifts of the stronger components from the center of subimage

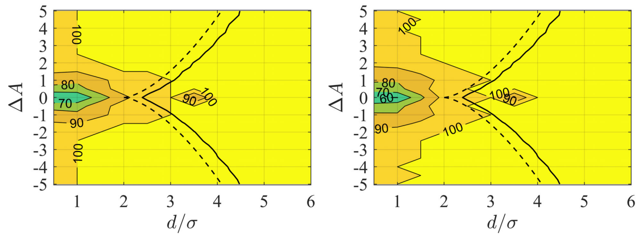
(**b**) Distance of the centres of weaker components to the nearest integers

FIGURE 5. **Distribution of centres of the profiles with respect to the nearest integers.**

positioned on the given distance d. The stronger component was positioned in the center of the sub-image but fractionally shifted within ± 0.5 pixels in both directions. The weaker component was positioned on the given distance d to the first one but at a random angular position. The random shifts were introduced to account for the random fractional sampling of the 2D Gaussian profile over a regular grid, as Fig. 5 shows. The center position of the weaker component also contains the integer and the fractional parts due to the fractional shifts of the stronger component, but also due to its random angular shift. Fig. 5 shows that the distribution of profiles' centers to the nearest integers is uniform for both components. Through the described procedure, the chosen sub-images are good representatives of profiles' sampling occurring in actual images.

The objects were iteratively extracted and removed based on the maximum image value. The samples of the highest intensities were sequentially declared as centers of the 2D Gaussian profiles and the ROIs were selected as circular regions relative to the widths of profiles.

The results of the IRWLS and LS methods are compared using the same scalar accuracy metric (15) for a joint evaluation of modeling gain. After the estimation and the removal of all 384 objects, the modeling errors for each sub-image using the IRWLS and LS methods were calculated and the



(**a**) LS method, the percentage of found objects

(**b**) IRWLS method, the percentage of found objects

FIGURE 6. **The percentage of found objects using the IRWLS and LS methods in $d/\sigma\,\Delta A$-plane defined by (5) and (6).**

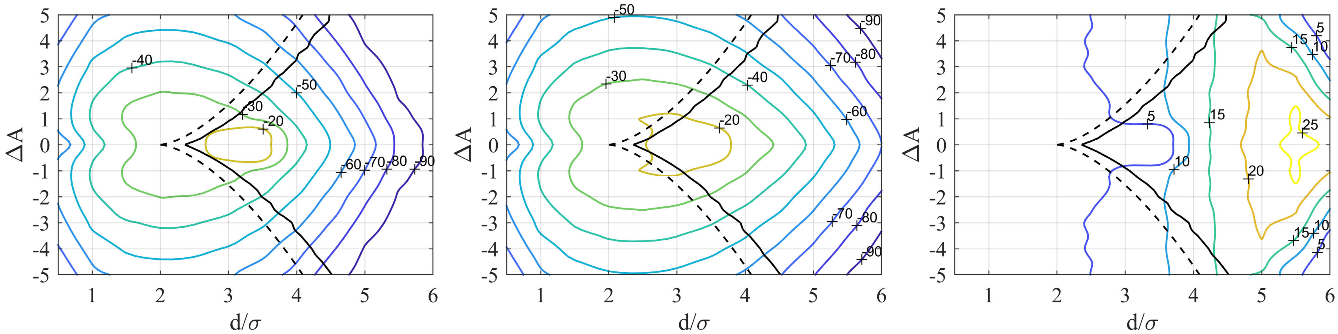
results are shown in Fig. 7. The contours, which represent the isolines of the same modeling error, were derived from uniformly sampled pairs from $d/\sigma\,\Delta A$-plane. For each pair, the median of modeling errors of sub-images was used to predict the accuracy. The results show that the IRWLS method achieves greater accuracy by up to 25 dB for certain combinations of contributing factors, and it is never worse than the LS solution. For example, for $(d/\sigma, \Delta A) = (5.5, 1)$ the IRWLS method achieves the accuracy of -85 dB, while the LS method achieves the accuracy of only -60 dB. According to the three-sigma-rule, 99.73% of the Gaussian profile is contained within $\pm 3\sigma$. Therefore, the analyzed center-to-center distance in the form of d/σ was chosen in the range from 0 to 6.

On the distance $d/\sigma = 0$, objects are completely overlapped and cannot be separated by either method but are modeled perfectly as a single object. On the other side, the circular profiles of the same widths are almost completely separated on distances $d/\sigma \geq 6$, again resulting in high accuracy for both methods. The results in Fig. 7 show that on small distances (e.g., $d/\sigma = 0.5$), both methods have similar relatively high accuracy (-60 dB) but estimate only one object instead of two. Such a compromise solution is subtracted from the synthesized image, thus producing a small modeling error due to a very small displacement of profiles. However, both methods failed in an attempt to separate objects on such small distances.

For the medium range of distances $1 < d/\sigma < 4$, the modeling error increases because many models were discarded due to certain rejection criteria (e.g., the estimated semiaxis λ_1 was too large, the determined ROI contained a too small number of samples, etc.). The invalid models were discarded and were not subtracted from the synthesized image, thus producing an increased modeling error in this region.

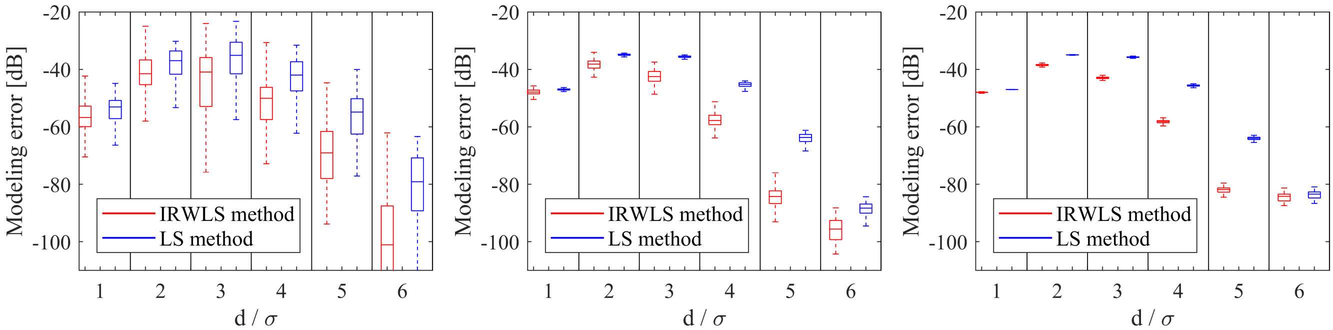
The percentages of found objects for the IRWLS and LS methods in the $d/\sigma\,\Delta A$-plane are shown in Fig. 6.

The results are similar for both methods. In an area where distances d/σ and differences of magnitudes ΔA are greater than 2, both methods find 100% of objects. Although it seems that the LS method finds the 100% of objects, the resolution limits, shown in Fig. 6, should be taken into account. The dashed black line represents the theoretical resolution limit in the continuous case according to (5) and (6), while the solid black line represents the empirical resolution limit in



(a) IRWLS method, modeling errors in dB **(b)** LS method, modeling errors in dB **(c)** IRWLS method modeling gain in dB

FIGURE 7. The modeling errors of the IRWLS and LS methods and the IRWLS method modeling gain in dB in $d/\sigma\,\Delta A$-plane. $\sigma = 1$.



(a) $\sigma = 0.5, \Delta A = 3$ **(b)** $\sigma = 1, \Delta A = 3$ **(c)** $\sigma = 2, \Delta A = 3$

FIGURE 8. Statistical analysis of modeling error for different profile widths σ.

the discrete case with 50% probability that a local minimum exists between profiles. In such a case, after the removal of the first compromise model, the LS method is trying to model the secondary object from the ridges which are residuals of this compromise model. In certain cases, such false models still fulfill the chosen validity criterion and are kept as valid secondary objects. On the other hand, with the IRWLS method, both estimated objects are valid since the method is based on the analysis of the residual errors, and not on the existence of a minimum between overlapped profiles. Furthermore, the IRWLS method gain is visible even in this unresolved area, as it is shown in Fig. 7c. It can be seen that the IRWLS modeling gain primarily depends on the center-to-center distance d/σ.

Additionally, in this region of medium distances, the accuracy also depends on differences in magnitudes. Namely, the gain of the IRWLS method is smaller when the objects have comparable magnitudes. In the case of equal magnitudes ($\Delta A = 0$), the error histograms of both components are overlapped and the boundary that would separate the samples cannot be drawn, thus making this method inapplicable for this special case. With sufficient magnitude differences (e.g. $\Delta A \geq 1$), the IRWLS method yields at least 5 dB improvement provided that profiles are separated by at least 2.5σ.

For the large range of distances $4 < d/\sigma < 6$, the IRWLS method with Huber weights demonstrates the highest modeling gain resulting from the precise estimation of profile parameters as it is shown on the right-hand side of Fig. 7c. On the distances $d/\sigma > 6$, the modeling gain of the IRWLS method is not significant, and both methods have the same efficiency.

B. THE INFLUENCE OF THE PROFILE WIDTH

The accuracy of the method also depends on the ratio of the profile width and the pixel size. Due to an iterative optimization process with six unknowns, the estimation accuracy depends on the number of input samples. The number of samples has to be greater or at least equal to the number of unknowns. A high ratio of the profile width and the pixel size yields a sufficient number of samples, thus providing higher accuracy and smaller parameter variances. The same Monte Carlo simulation was performed for different profile widths assuming circular objects of same widths of semiaxes $\lambda_1 = \lambda_2 = \sigma$ for $\sigma \in \{0.5, 1, 2\}pix$ with the equivalent full width at half maximum (FWHM) of $\cong 2.35\sigma$.

The statistical analysis of modeling error is shown in Fig. 8. By selecting $\sigma = 0.5\ pix$, the effective profile area is reduced four times, while by selecting $\sigma = 2\ pix$, the profile area is increased by the same factor. The results show that the selection of $\sigma < 1\ pix$ increases the standard deviations of errors and slightly reduces the median of modeling accuracy. For $\sigma = 1\ pix$ and $\sigma = 2pix$, the results are similar. In both cases, the objects are large enough and contain a sufficient

TABLE 1. The execution time required for the estimation of one 2D Gaussian profile. The average number of input samples per profile was 29, the average number of outer loop iterations was 20 for the center-to-center distance $d/\sigma = 4$.

	With analytical derivatives	Without analytical derivatives
Mean time [s]	0.1591	0.3665
Median time [s]	0.1313	0.3137
Maximum time [s]	0.5027	1.0865
Minimum time [s]	0.0438	0.1125

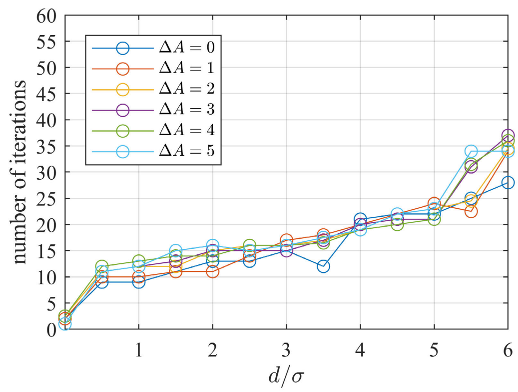


FIGURE 9. Medial number of outer loop iterations for σ =1 *pix*, *tol* = 10^{-5}, different center-to-center distances d/σ, and different ΔA.

number of samples resulting in an overdetermined optimization problem. The modeling error and parameters' variances are also decreased with the increase of the profile width and the number of input samples.

C. THE SPEED OF THE ALGORITHM

The analysis of the algorithm speed is performed by measuring the median number of outer loop iterations for the estimation of each of 384 objects. The chosen stopping criterion for the outer loop is that the relative change of threshold δ is smaller than 10^{-5}. The updated threshold δ in each outer loop iteration is used for the calculation of weights. The updated weights pass to the objective function enabling the optimization process to start a new inner loop iteration. The termination tolerance on the objective function value and the current point were set to 10^{-10}. Also, the maximal number of outer loop iterations was unlimited. The complexity of the algorithm was calculated by measuring the elapsed time required for the estimation of one of 384 profiles. The experiment was executed on a system with Intel(R) Core(TM) i5-7200U CPU @ 2.5 GHz and 8 GB of RAM. The optimization was performed using the native MATLAB fminunc solver. The execution time statistics is shown in Table 1. If analytic derivatives are supplied, the mean estimation time is 0.1591 seconds per profile. Without supplied derivatives, the mean estimation time is increased to 0.3665 seconds per profile, thus showing that the analytic derivatives accelerate the estimation process more than two times.

Fig. 9 shows the medial number of outer loop iterations for different values of the center-to-center distances d/σ and differences of profiles' magnitudes ΔA. With the larger center-to-center distance d/σ, the contribution of the IRWLS method is increased, but so is the number of required iterations. By limiting the maximum number of the outer loop iterations t_{max} to only five, about 80% of the maximum improvement of the IRWLS method compared to the LS solution is still achieved, thus showing that the method is useful even in the applications requiring small complexity.

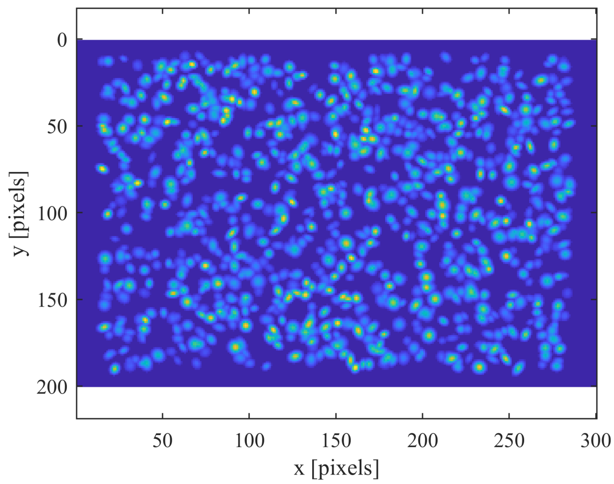


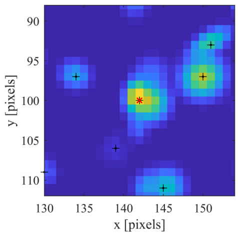
FIGURE 10. Synthetic image with 1000 random objects (due to the high dynamic range, $z^{1/4}$ is shown).

D. SYNTHETIC RANDOM IMAGE

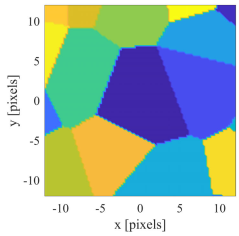
The second experiment simulates the estimation of parameters in actual astronomical images. The parameters of the profiles were estimated from the synthetic image that contained 1000 objects of random magnitudes, center positions, semi-axes widths, and rotation angles, independently and randomly chosen for each object. The image size was 200 × 300 pixels including the padding of 10% (Fig.10).

In the previous experiment, we demonstrated that the IRWLS method achieves a gain of up to 25 dB at the center-to-center distances $d/\sigma > 4$. The question naturally arises as to why we did not segment the ROI before the estimation process at such large distances, thus reducing the error of the LS method, and whether the IRWLS method would still have a smaller modeling error than the LS method. Therefore, the tessellation of the ROI was also performed in this experiment, and the results show that, even after the segmentation, the IRWLS method still has a substantial modeling gain.

The segmentation assumes the determination of affiliation of each sample within the ROI to one of the adjacent profiles. The segmentation criterion used in this experiment represents a generalized Voronoi tessellation. Instead of the ordinary Euclidean distance, a generalized distance, which combines the squared Euclidean distance with the additional offset term determined by the differences of magnitudes of the adjacent objects, is used as the criterion of closeness. Generally, the affiliation of the sample to a certain profile includes the search for generalized bisectors between each pair of neighboring profiles and the intersection of those bisectors, thus forming convex areas of profiles' affiliations. The chosen

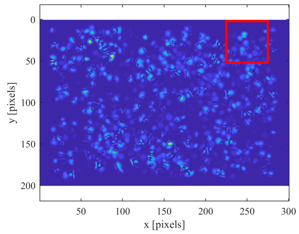


(a) The region around object selected for estimation

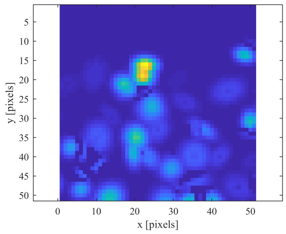


(b) Tessellation around selected object

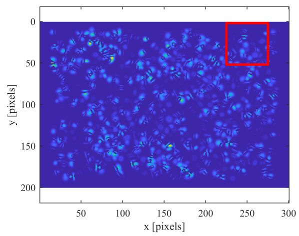
FIGURE 11. **The example of generalized linear tessellation.**



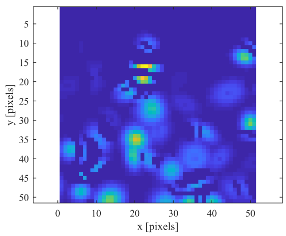
(a) Residual error of the IRWLS method with tessellation



(b) Residual error of the IRWLS method (zoomed red square)

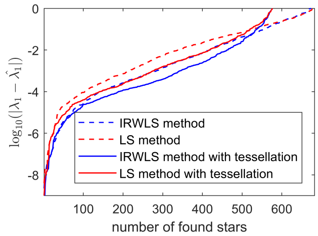


(c) Residual error of the LS method with tessellation

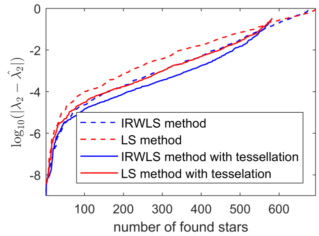


(d) Residual error of the LS method (zoomed red square)

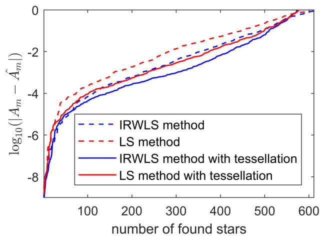
FIGURE 12. **Residual errors obtained by the IRWLS and LS methods with tessellation (to emphasize small error values, $|\mathbf{e}|^{1/4}$ is shown).**



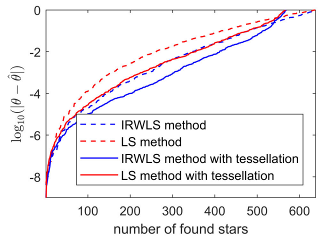
(a) Sorted log10-semiaxis λ_1 errors



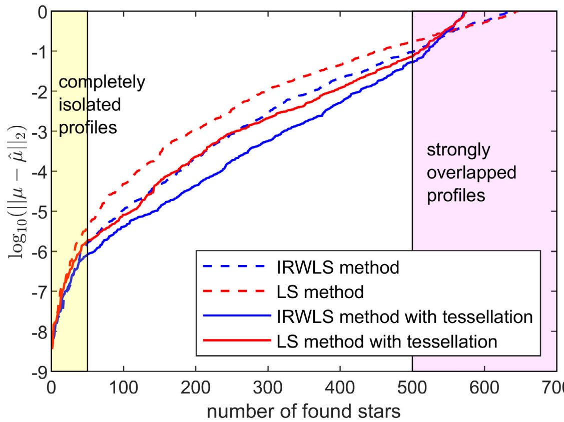
(b) Sorted log10-semiaxis λ_2 errors



(c) Sorted log10-magnitude A errors



(d) Sorted log10-angle θ errors



(e) Sorted log10-distances of estimated and actual center positions

FIGURE 13. **Sorted log10-absolute errors of estimated parameters for two object extraction methods: 1) iterative object removal based on image maximum value (dashed lines) and 2) sorted list of local extrema of the blurred image with generalized linear tessellation (solid lines).**

segmentation criterion makes the ROIs of stronger components larger at the expense of the weaker components following the absolute magnitudes' difference. The example of the applied segmentation is shown in Fig. 11.

Although the tessellation assumed only one object in the central cell of the tessellated region, in this example, it is obvious that there are more than one overlapped objects which are too close to be separated. For such a case with only one local maximum, the IRWLS method can offer a clear advantage.

Since the object positions are generally unknown, the estimation of the ROI begins with the method of local extrema. The samples of the highest intensities are sequentially declared as potential candidates for centers of the 2D Gaussian profiles since the synthesized image is noiseless. In the case of the actual astronomical images, the images should be pre-blurred to ensure that isolated wrong pixels or pixels contaminated by noise would not be declared as extrema. The first and second moments were used to find the initial parameters for the optimization procedure.

The ROI was determined in two ways: 1) relative to the initial center position and the expected widths of profile semiaxes without tessellation (e.g. $\pm 3\sigma$ around the initial center position) or 2) using the described segmentation procedure. After the estimation process, each valid estimated model is subtracted from the whole synthetic image while the invalid models are skipped. The procedure of profile estimation and removal was iteratively repeated in accordance with the matching pursuit concept. The comparison of residual errors obtained by the application of the IRWLS and LS methods using the generalized Voronoi tessellation is shown in Fig. 12.

Although the modeling residuals in Fig. 12a and Fig. 12c look quite similar, subtle differences can be observed in the zoomed regions. It can be seen that the LS method produced a larger number of false ridges than the IRWLS method and consequently the worse estimate of parameters. The sorted log10-errors of the estimated parameters with tessellation (solid lines) and without tessellation (dashed lines) are shown in Fig. 13. Absolute errors of semiaxes lengths λ_1, λ_2, the rotation angle θ, the magnitude A, as well as the distance of the actual and estimated center position $d = \sqrt{(\mu_x - \hat{\mu_x})^2 + (\mu_y - \hat{\mu_y})^2}$ are given. The results show that the IRWLS method yields a more accurate estimation of parameters by up to one order of magnitude even when the segmentation of ROI is applied.

The IRWLS method yields more accurate estimates than the LS method for a significant percentage of objects in the central zone of Fig. 13e. All parameters estimated using the IRWLS method have consistent improvements as regards to the LS method.

VI. FUTURE WORK

The preliminary research, published in [34], investigated the influence of the Huber estimator tuning constant on the convergence rate and the efficiency of the IRWLS algorithm. As expected, the tuning constant in the relatively wide range from 1 to 2 results in approximately the same IRWLS method efficiency. However, it primarily affects the convergence rate where the smaller values of the tuning constant yield a solution closer to the L_1 norm with a slower convergence rate, while the greater values of the tuning constant yield a solution closer to the L_2 norm with the faster convergence rate.

The future work will be focused on the verification of the proposed method to actual astronomical images. The other extension will also be an analysis of the influence of different types of noise on the parameter variances and efficiency of the robust method. It is expected that the robust method should also give a clear advantage in noisy images with or without overlap.

VII. CONCLUSION

The main advantage of the proposed robust iterative method compared to the least squares method is the more precise estimation of Gaussian profile parameters, when neither the centers of the profiles nor their total number are known. Since the method is based only on the histogram of residual errors, it recognizes the outliers and applies soft thresholding, which with an appropriate threshold δ provides unbiased and efficient estimates as it is demonstrated through Monte Carlo analysis. It was shown that the method achieves greater accuracy up to 25 dB compared to the conventional least squares method for a certain set of differences of profiles' magnitudes and center-to-center distances. Moreover, the 80% of total improvement is achieved in only five iterations. As expected, the modeling gain is primarily determined by the distances of profiles' centers. For the case of rotationally symmetric profiles, the minimum ratio of the effective profile width and the pixel size should be around one in order to have a sufficient number of samples for the optimization process and smaller variances of the estimated parameters. Through the synthetic example, we demonstrated the applicability of the proposed method for the analysis of astronomical images.

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